



Girijananda Chowdhury University Guwahati 781017, Assam

UNIT TEST - II

BCE23405T Structural Analysis

Session: 2025-2026

Date: 22 May 2026

Time: 1 hour

Batch: 2024-2028

Total Marks: 20

Question 1: Fill in the Blanks (5 marks)

[CO4]

- The stiffness factor of a beam fixed at one end and hinged at the other end is _____
- The slope deflection method is a _____ method of structural analysis.
- The fixed end moment for a beam carrying a uniformly distributed load over the entire span is _____ at both ends.
- The process of transferring half of the unbalanced moment to the far end is called _____.
- A joint is said to be balanced when the algebraic sum of moments at the joint becomes _____.

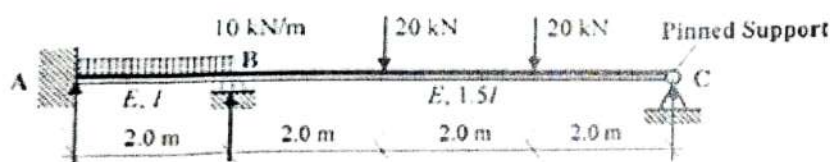
Question 2 (8 marks)

[CO4]

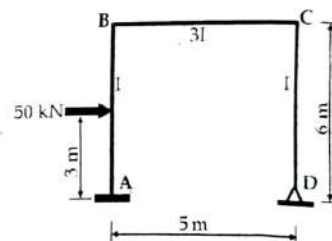
A) Analyze the continuous beam (Figure 2A) using the **slope deflection method** and determine the end moments, if support B sinks by 20 mm. Draw Bending Moment Diagrams. Take EI as $1 \times 10^5 \text{ kN-m}^2$

OR

B) Analyze the Frame (Figure 2B) using the slope deflection method and determine the end moments. Draw Bending Moment Diagrams.



(Figure 2A)



(Figure 2B)

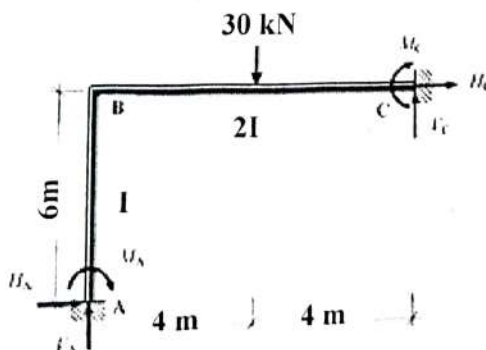
Question 3 (7 marks)

[CO4]

A) Analyze a two-span continuous beam ABC with fixed support at A and hinged at C. Span AB = 5 m carries a UDL of 16 kN/m and span BC = 6 m carries a point load of 70 kN at one-third span from B. Determine the support moments using **Moment distribution method** and sketch the bending moment diagram.

OR

B) Analyze the Frame (Figure 3B) using **Moment distribution method** and determine the end moments. Draw Bending Moment Diagrams.



(Figure 3B)